

Binding activities to a service.

Application components (clients) can bind to a Service by calling `bindService()`. The Android system then calls the Service's `onBind()` method, which returns an `IBinder` for interacting with the Service. The binding is asynchronous.

Types of Android Services.

Foreground Services:

Foreground Services are those services that are visible to the users.

Background Services:

These services run in the background, such that the user can't see or access them.

Bound Services :

Bound Service runs as long as some other application components is bound to it.

All the Service that you have seen are simple - either they start with a counter and increment at regular intervals or they download a fixed set of files from the internet.

1. First, in the calling activity you create an intent object, specifying the service name:

```
Public void StartService (view view) {  
    Intent Intent = new Intent (getBaseContext().  
        MyService .class);
```

2. you then create an array of URL objects and assign it to the Intent object through its putExtra() method.
3. you start the service using the intent object:

```

Public void startService (view view) {
    Intent intent = new Intent (getBaseContext(), MyService.class);
    try {
        URL[] urls = new URL[] {
            new URL ("http://www.amazon.com/Somefiles.pdf"),
            new URL ("http://www.wrox.com/Somefiles.pdf"),
            new URL ("http://www.google.com/Somefiles.pdf"),
            new URL ("http://www.learn2develop.net/Somefiles.pdf")
        };
        intent.putExtra ("URLS", urls);
    } catch (MalformedURLException e) {
        e.printStackTrace();
    }
    startService (intent);
}

```

4. Note that the URL array is assigned to the intent object as an object array.

5. On the service's end, you need to extract the data passed in through the Intent object in the onStartCommand() method:

```

@Override
Public int onStartCommand (Intent intent, int flags, int startId) {
    Toast.makeText (this, "Service started", Toast.LENGTH_LONG).show();
    Object[] objUrls = (Object[]) intent.getExtras().get ("URLS");
}

```

```

URL[] urls = new URL[ objUrls.length ];
for (int i=0; i<objUrls.length-1; i++) {
    urls[i] = (URL) objUrls[i];
}
new DoBackgroundTask().execute(urls);
return START_STICKY;
}

```

6. The preceding `findViewById()` extracts the data using the `getExtras()` method to return a `Bundle` object.

7. It then uses the `get()` method to extract the URL array as an object array.

8. Because in Java you cannot directly cast an array from one type to another, you must create a loop and cast each member of the array individually.

9. Finally you execute the background task by passing the URL array into the `execute()` method.